Syeda Sumayya Zahid

Software Engineer | Robotic And AI Enthusiast

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OBJECTIVE

Software Engineer with hands-on experience across full-stack development, applied AI/ML, and robotics, seeking to join an engineering team where I can build production-grade AI systems from LLM-powered applications and RAG pipelines to intelligent, real-time robotics and grow into a senior AI engineering role.

EXPERIENCE

AI Engineer | Arbotrix

Onsite | March 2026 – Pesent

- Developed and trained ML/DL models (CNNs, neural networks) for intelligent robotics and AI-driven production applications.
- Engineered ROS 2-based robotic software systems covering node communication, sensor integration, and autonomous operations.
- Designed and deployed full-stack web applications with RESTful APIs, responsive frontends, and scalable backend architectures.
- Integrated trained AI models into live robotic workflows enabling real-time decision-making and adaptive system behaviour.
- Performed end-to-end dataset preprocessing, model evaluation, hyperparameter tuning, and performance optimisation.
- Contributed to robotics software debugging, hardware-software interfacing, and system-level testing.
- Collaborated with cross-functional engineering teams to architect and deliver scalable AI-powered robotics solutions.
- Participated in R&D initiatives for next-generation AI, automation, and autonomous robotics technologies.

AI Model trainer | TechKnock.tech

Remote | March 2025 – Feb 2026

- Trained and optimised ML and DL models targeting high accuracy and low-latency real-time inference.
- Designed and managed end-to-end data pipelines: collection, cleaning, augmentation, and versioning for reproducible experiments.
- Conducted rigorous model evaluation and validation cycles to benchmark and improve reliability across diverse use cases.
- Collaborated with autonomous systems teams to integrate trained models into robotic and software application workflows.

PROJECTS

DODO Robot — *Role: AI and Robotic engineer*

- Integrated AI models into robotic workflows for automation and decision-making.
- Assisted in sensor handling, robot communication, and control system implementation.
- Participated in robotic system testing, debugging, and simulation.

- Contributed to AI model training and performance optimization for robotics applications.

RAGspace - Retrieval-Augmented Generation Platform — *Role: AI Engineer*

- Built a full-stack RAG application (Next.js, FastAPI, Docker) enabling document Q&A grounded in source citations.
- Implemented a local-first RAG pipeline using LangChain, ChromaDB, and Ollama for embeddings and inference.
- Designed document chunking and vector embedding generation using nomic-embed-text for semantic search.

LLM Cost-Latency Router — *Role: AI engineer*

- Built an ML-based prompt classifier (TF-IDF + Scikit-Learn) to route requests by complexity across LLMs.
- Reduced inference cost and latency by dispatching simple prompts to smaller models and complex ones to larger models.
- Integrated multi-provider LLM routing via LiteLLM, supporting OpenAI, Groq, Anthropic, and more.
- Built a real-time Streamlit analytics dashboard tracking cost, latency, and model usage from SQLite logs

AI Learn Hub - AI -powered Online platform — *Role: Full stack developer*

- Architected a TypeScript full-stack application (React + Node.js + Firebase) with AI integration using Claude API for dynamic quiz generation from course content.
- Implemented Firebase Authentication with custom JWT tokens, role-based middleware (Student/Instructor/Admin), and secure session management.
- Built a RESTful API handling course CRUD, lesson management, enrollment flow, AI quiz generation/submission, and automated grading with weak-topic identification.
- Developed adaptive learning features including performance-based recommendations, progress tracking, and personalized result analytics.
- Designed the frontend with component-based architecture using React Router, Context API, and Tailwind CSS, ensuring responsive design across devices.

Arbotrix - Company and Portfolio Website — *Role: Full stack developer*

- Designed and developed the official company website for Arbotrix.tech using Next.js, Node.js, and Tailwind CSS.
- Built a server-side rendered frontend with Next.js for fast load times, SEO optimisation, and dynamic routing.
- Implemented responsive UI components using Tailwind CSS ensuring a polished, professional experience across all devices.
- Developed backend logic and API routes with Node.js for dynamic content handling and efficient server-side processing.
- Managed full project version control using Git/GitHub including branching, commits, and production deployment pushes.
- Optimised performance, cross-browser compatibility, and mobile responsiveness aligned with the company's technical branding.

AI-Powered Emergency Response System — *Role: AI and System Development Final Year Project*

- Developed an AI-powered emergency response application capable of detecting threats using voice emotion recognition and movement analysis.
- Integrated intelligent detection mechanisms for real-time emergency identification and alert generation.
- Worked on AI-based analysis techniques to improve response accuracy and public safety assistance.
- Contributed to system testing, documentation, and performance evaluation.

AI-Powered Skin Cancer Detection — *Role: AI/ML and Full stack developer*

- Developed a full-stack AI-powered skin cancer detection system using Next.js, FastAPI, TensorFlow, ResNet50, and SQLite.
- Trained and optimized a ResNet50 deep learning model on the HAM10000 dataset to classify 7 skin lesion types with high accuracy.

- Built secure REST APIs with JWT authentication, integrated OpenCV for image preprocessing/validation, and managed user data with SQLAlchemy and SQLite.
- Designed a responsive interface for real-time image analysis, prediction confidence scores, scan history, and medical recommendations using Next.js and Tailwind CSS.

Other Projects: Cinematic-Roaster-App (full-stack AI web app detecting negative movie reviews and generating sarcastic AI roasts using chained Hugging Face models — Django, Next.js, PyTorch) · GridCore360 (enterprise AI & BPO platform — Next.js, TypeScript, Firebase, Tailwind, Framer Motion) · Portfolio (personal site — Next.js, TypeScript, Tailwind CSS, Framer Motion) · RS Beauty Saloon (booking & admin platform, Next.js/Node.js) · Tera X Autonomous Robot (hardware design and sensor–ROS 2 interfacing) · Emotion Detection (CNN-based facial emotion classification) · Al Rehman Welfare App (donation, volunteer & event management platform).

EDUCATION

BS Software Engineering

Comsats University Islamabad,Abbottabad Campus | 2022 – 2026 | Grade : A+

FSc (Pre-Engineering)

Islamia College, Peshware | 2019 – 2021 | Grade: A+

TOOLS & SKILLS

AI & Machine Learning: Python, Machine Learning, Deep Learning, TensorFlow, Keras, Neural Networks, CNNs, RAG,Transfer Learning, Model Training & Optimisation, Hyperparameter Tuning, Data Preprocessing & Augmentation, AI Model Evaluation

Computer Vision: OpenCV, Object Detection, Image Classification, Facial Expression Recognition, Real-Time Video Processing

Robotics & Autonomous Systems: ROS 2, Autonomous Systems Development, Sensor Integration & Fusion, Hardware-Software Interfacing, Robot Simulation, Robotics Control Systems, AI for Robotics

Web Development: Next.js, Node.js, Tailwind CSS, HTML5, CSS3, JavaScript, RESTful API Development, Responsive Design, Database Integration, Server-Side Rendering

Software Engineering & QA: SDLC, Agile / Scrum Methodologies, API Testing, Manual Testing, Debugging, Requirement Analysis, Version Control (Git/GitHub)

Platforms & Tools: Google Colab, VS Code, Postman, Jira, GitHub

Communication & Collaboration: Slack, Microsoft Teams, Zoom, Confluence

Business & Productivity: Google Sheets, Microsoft Excel